FRONT AXLE - 181FBI (Continued)

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has been adjusted correctly. The backlash can be adjusted within specifications to achieve desired tooth contact patterns.

The TOP LAND of the gear tooth is the top surface of the tooth. The PROFILE of the gear tooth is the depth of the tooth. The TOE of the gear is the portion of the tooth surface at the end towards the center. The HEEL of the gear is the portion of the tooth at the outer-end. The ROOT of the gear tooth is the lowest portion of the tooth (Fig. 25).

NOTE: If the PROFILE across the tooth is the same it is a 3 Axis cut gear. If the PROFILE across the tooth is tapered it is a 2 Axis cut gear.

(1) Apply a thin coat of hydrated ferric oxide or equivalent to the drive and coast side of the ring gear teeth.

(2) Wrap, twist and hold a shop towel around the pinion yoke to increase the turning resistance of the pinion. This will provide a more distinct contact pattern.

(3) With a boxed end wrench on the ring gear bolt, rotate the differential case one complete revolution in both directions while a load is being applied from shop towel.

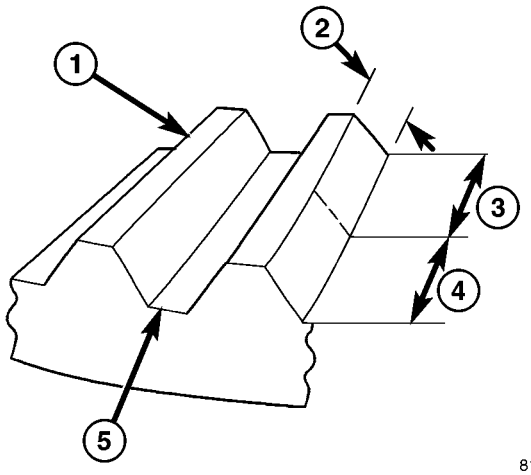

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Fig. 25 GEAR DESCRIPTION

1 - TOP LAND
2 - PROFILE
3 - TOE
4 - HEEL
5 - ROOT

The areas on the ring gear teeth with the greatest degree of contact against the pinion teeth will squeeze the compound to the areas with the least amount of contact. Note and compare patterns on the ring gear teeth to Gear Tooth Contact Patterns chart (Fig. 26) and (Fig. 27) and adjust pinion depth and gear backlash as necessary.

DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top heel contact		**Top toe contact**	Backlash correct. Thicker pinion position shim required.
	Root toe contact		**Root heel contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top heel contact**	Pinion position shim correct. Decrease backlash.
	Root toe contact		**Root toe contact**	Pinion position shim correct. Increase backlash.

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Fig. 26 PATTERN INTERPRETATION (GEAR CUT 2 AXIS)